

Autumn Semester Examination-2022

Subject: Engineering Physics

Subject Code: PHYUGBS01

Full Marks: 80

Time: 3 hrs

▪ Answer any eight (8) questions from the following.

- 1) (a). If $\mathbf{A} = 2\mathbf{i} + 2\mathbf{j} - \mathbf{k}$ and $\mathbf{B} = 6\mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$, find the angle between $\mathbf{A}$ and $\mathbf{B}$ vectors.
 (b) Show that the vectors $\mathbf{A} = \mathbf{i} + 4\mathbf{j} + 3\mathbf{k}$ and $\mathbf{B} = 4\mathbf{i} + 2\mathbf{j} - 4\mathbf{k}$ are mutually perpendicular.
 (c) On application of a force $5\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$ N, a particle moves from position $2\mathbf{i} + 5\mathbf{j} + 5\mathbf{k}$ m to another position $12\mathbf{i} + 15\mathbf{j} + 8\mathbf{k}$. Calculate the work done.
 (d) Find the area of the parallelogram determined by the vectors $\mathbf{A} = 3\mathbf{i} + 2\mathbf{j}$ and $\mathbf{B} = 2\mathbf{j} - 4\mathbf{k}$.
 (e) If $|\mathbf{P}| = 10$, $|\mathbf{Q}| = 1$ and $\mathbf{P} \cdot \mathbf{Q} = 6$, obtain $|\mathbf{P} \times \mathbf{Q}|$. (2x5=10)
- 2) (a) Write Coulomb's law. (b) Write down the properties of electric lines of force.
 (c) Write Gauss's law and express it mathematically. (d) What is an Electric Dipole?
 (e) Calculate the torque on an electric dipole in a uniform electric field. (2x5=10)
- 3) What do you mean by central force? Give some examples of central force. Write down the characteristics of central force. What is the areal velocity. What is the physical significance of cross product? (2 + 1 + 3 + 2 + 2)
- 4) What is Stoke's law and write its equation. Write the difference between the streamline flow and laminar flow. Define equation of continuity. What is Reynolds number and what is the importance of it? What is Bernoulli theorem and its applications? (2 + 3 + 1 + 2 + 2)
- 5) (a) What is Photo electric effect? Explain the effect of increase of incident radiation on photo electric current with suitable graph. Which metal exhibit photo electric effect? (b) State the Heisenberg Uncertainty principle. Write the formula in terms of momentum and position of a system. (c) What is the significance of Ψ in Schrodinger wave equation? (2+2+2+2+1+1)
- 6) (a) Explain the terms binding energy of a nucleus and the binding energy per nucleon. Show graphically the variation of binding energy per nucleon with mass number. (b) Write the relation between mass number and radius of the nucleus. (c) What is a nuclear reactor? (d) Write the uses of radio isotopes. (2+2+2+1+1+2)
- 7) (a) What is the need for doping in semiconductor? (b) Describe n-type and p-type semiconductors. What are the majority and minority charge carriers in each type of semiconductor? (c) Draw the I-V characteristics curve of a P-N junction in forward bias. (2+2+2+2+2)
- 8) (a) Derive Bragg's law of crystal diffraction. (b) If X-ray of wavelength $100 \text{ \AA}$ is incident on an atom at an angle of 90° , then what should be the value of d for first order spectrum? (c) What is the formula for Mosley's law? (d) Write down the comparison between Dia-magnetism, para-magnetism and ferro-magnetism. (3+2+2+3)
- 9) (a) Derive the expression for the fringe width in a Young's double slit experiment. How will the fringe width change if (i) separation between the slits is increased (ii) screen is moved away from the plane of the slits. (b) Define 'Fringes of equal thickness' (FET). Give an example
 (c) Is it possible to have interference without diffraction? (3+1+1)+ (2+1)+ (2)
- 10) (a) Define H-polaroid.
 (b) An EM wave $E_x(z, t) = E_{0x} \cos(kz - \omega t)$ $E_y(z, t) = E_{0y} \cos(kz - \omega t + \delta)$
 Find the state of polarization in following cases (i) $\delta = 0, E_{0x} \neq E_{0y}$; (ii) $\delta = \frac{\pi}{2}, E_{0x} = E_{0y}$;
 (c) What are the difference between the Fraunhofer diffraction and the Fresnel diffraction?
 (d) Define Laevo-rotatory substance. (2+(2+2)+2+2)

f = 6\pi n\tau O

Z = p r e